



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECP3843
SPECIAL TOPIC IN DATA ENGINEERING

SECTION 02
GROUP 27

ASSIGNMENT 1 (TECHNICAL REPORT)
TITLE: STREAMING DATA & BATCH PROCESSING

LECTURER:
DR. ARYATI BINTI BAKRI

PREPARED BY:
EVELYN GOH YUAN QI (A23CS0222)
JOANNE CHING YIN XUAN (A23CS0227)

EXECUTIVE SUMMARY

This report examines batch and streaming data processing in enterprise systems. It discusses their operations, applications, advantages, and disadvantages. Most data comes from digital platforms, enterprise applications, and IOT devices. Therefore, organizations find it challenging to handle the high volume and high velocity of data streams very effectively. The less time-sensitive tasks such as inventory reporting, financial statement, and strategic planning are still best done with batch processing that processes large datasets at some scheduled time interval. It is simple and low-cost, but it cannot be very responsive. Streaming data processing allows real-time processing of applications that need immediate response like fraud detection, real-time e-commerce recommendations and IOT monitoring. However, it is quite complex and requires systems that are under constant monitoring. System complexity, operation instability, and data heterogeneity are the main issues. The report discusses unified Kappa architecture, self-regulating framework and robust data integration as the ways of addressing these challenges. These approaches allow enterprises to efficiently process both historical and real-time data, thereby helping them make important and timely decisions.

1.0 INTRODUCTION

Today's digital platforms, enterprise applications, and IoT devices continuously generate data in massive volumes, which results in difficulties in managing decision-making and business processes. Companies require scalable data processing solutions in order to efficiently manage large-scale and fast-moving data streams (Hashem et al., 2015).

Two major approaches have emerged to tackling these problems: batch and streaming processing. Batch processing refers to handling a complete dataset in batches over time. In this case, the processing is slower.

On the other hand, streaming data processing allows organizations to handle a large amount of live data and respond to events immediately (Kleppmann, 2017). For example, banks employ streaming data processing for finding fraudulent activities and monitoring transactions in real-time, while enterprise reporting and business intelligence systems rely on batch processing to generate quality analytical information (Chen et al., 2014).

This paper elaborates on and contrasts batch processing and streaming data processing from the perspective of their functionality, advantages, and disadvantages in business environments. The discussion centers on the enterprise level and conceptual analysis, leaving out the details of technical implementation. The goal is to assess how these data processing paradigms facilitate effective data management and support decision-making in up-to-date organizations.

2.0 BACKGROUND / LITERATURE REVIEW

The shift to high-performance data systems is largely driven by the worldwide data explosion recognized as the global Data Sphere, which is expected to reach 175 Zettabytes by 2025 (Benjelloun et al., 2020). Traditionally, companies depended on batch processing which in simple terms means that data is collected in large static datasets and then processed later. Essentially, this batch processing method built on the main architecture of standard Database Management Systems (DBMS) has given a static snapshot of data by carrying out one-off queries (Golab & Özsu, 2003). Yet, as the data volumes keep surpassing the boundaries of the conventional ways of scaling, batch processing in its traditional form is viewed as being very slow and is turning into a bottleneck. And so stream processing, with its continuous, high-volume data arriving and being processed on the fly by persistently running queries that offer real-time insights, has gained popularity (Koganti et al., 2025).

Due to these capabilities being unified into enterprise systems, the literature shows that architectures have evolved from Lambda to Kappa. Lambda architecture was first formulated to balance real-time "speed layers" with historical "batch layers," but it has been argued by researchers that this dual-pipeline approach raises the level of complexity and maintenance costs too much (Penka et al., 2021). Therefore, the Kappa architecture has been developed by considering all data as one stream, thus radically simplifying the system to a single processing engine (Penka et al., 2021). There are some modern frameworks that support this shift such as Apache Spark, which uses micro-batching to bridge the gap between the paradigms, and Apache Flink or Storm, provide real per-event processing for sub-second latency (Hesse & Lorenz, 2015). While batch processing is still a viable choice for conducting analytical tasks that do not require immediate results, current studies show that high-performing stream processing has become a key enabling technology of modern enterprise systems, especially in the field of IoT, finance, and smart agriculture (Koganti et al., 2025).

3.0 ANALYSIS AND DISCUSSION

3.1 BATCH PROCESSING ARCHITECTURE AND OPERATION

Batch processing is a structured, sequential model in which data is collected, stored, and processed at scheduled intervals. Figure 1 shows that the data is stored over time and processed in batches with outputs generated at the end of every processing cycle. The method emphasizes throughput and computational efficiency and can be applied in situations of large-scale data analysis. However, the latency of batch processing also exists because of the natural

delay between data processing and data collection, which is a constraint to the effectiveness of batch systems in time-sensitive applications (Chen et al., 2014).

3.2 STREAMING DATA PROCESSING ARCHITECTURE AND OPERATION

Streaming data processing, in contrast, is an event-driven architecture, meaning data is processed as it is created. This system, illustrated in Figure 1, receives and handles real-time information without any prior accumulations, which makes it possible to have immediate feedback and rapid response to changing conditions. Streaming can be extensively used with real-time applications, with technologies like Apache Kafka and Apache Flink able to handle low-latency processing in constantly changing environments (Kleppmann, 2017).

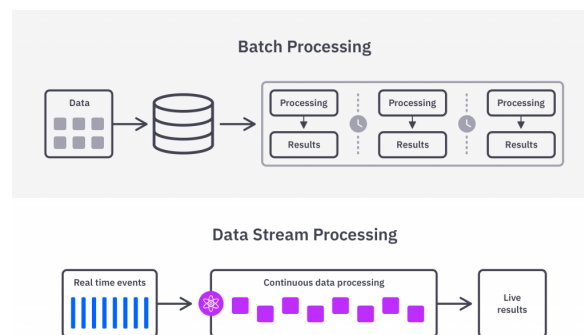


Figure 1. Batch processing and streaming data processing. Adapted from *Estuary* (2023).

3.3 CASE EXAMPLES OR APPLICATIONS

Batch processing is still necessary in the system of the enterprise when real-time processing is not a necessity. Walmart, a large organization, is an example that makes use of batch processing to compute reports on sales data, which is used to produce daily or weekly reports on inventory management and strategic planning. Similarly, the banking systems handle end-of-day transactions as batches in order to generate financial statements and maintain data coherence across systems (Gandomi & Haider, 2015).

Streaming data processing plays an important role in applications that demand real-time response. For instance, PayPal employs real-time data processing to eliminate fraudulent transactions through user behaviour analysis as the transactions occur. Amazon uses streaming data in the e-commerce industry to offer real-time product suggestions to customers based on their interactions. Also, Uber uses streaming data processing to dynamically pair drivers and passengers, aiming to find the best paths in real-time.

Streaming processing is applied in smart agriculture and industrial monitoring systems, in which sensor data is constantly processed to identify anomalies and assist in predictive maintenance in Internet of Things applications (Koganti et al., 2025).

3.4 ADVANTAGES AND DISADVANTAGES

Processing Type	Advantages	Disadvantages
Batch Processing	• High efficiency for large datasets • Lower operational cost • Simpler system design	• High latency • Not suitable for time-sensitive applications
Streaming Processing	• Real-time data processing • Enables rapid decision-making	• Higher system complexity • Increased infrastructure cost • Requires continuous monitoring

Table 1: Advantages and Disadvantages of Batch and Streaming Processing

These advantages and limitations highlight the trade-offs between batch and streaming processing, which contribute to the challenges discussed in the following section and justify the need for advanced architectural solutions.

3.5 CHALLENGES AND FUTURE TRENDS

While real-time data provides various benefits, organizations still face challenges when transitioning from traditional batch processing systems to streaming processing systems because these systems bring about new technical and operational challenges, as discussed earlier. Firstly, data heterogeneity continues to be a major challenge since it is difficult to integrate and process in real time various data formats coming from different sources (Özbulak et al., 2025). Another issue is that streaming necessitates sophisticated state management for accurate processing, so that it is actually more complex to maintain than batch systems (Golab & Özsu, 2003). Moreover, many systems suffer from operational failures as they are unable to accommodate sudden bursts of data even with manual intervention (Floratou & Agrawal, 2017).

In order to solve these challenges, the industry is moving toward more intelligent and automated solutions. Self-regulating systems employ machine learning to automatically adjust the scaling of resources, thereby minimizing manual intervention (Floratou & Agrawal, 2017). Stream reasoning facilitates performing real-time analytics of complex data, especially in IoT and smart city contexts (Özbulak et al., 2025). Furthermore, the adoption of Kappa architecture makes data processing easier by introducing one system only for both real-time and historical data (Penka et al., 2021).

4.0 RECOMMENDATIONS

To successfully shift from traditional batch processing to up-to-date streaming processing, organizations should adopt solutions that will help in directly handling the issues

discussed earlier. Firstly, one of the best ways for organizations to reduce the system complexity that comes from using Lambda architecture is to change it to a Kappa architecture that is unified. If both real-time and historical data are considered as a single data stream, then organizations won't have to maintain two separate codebases, which is usually the major cause of high maintenance costs and data inconsistencies (Penka et al., 2021). Such a move makes systems more reliable and also leads to faster deployment of new analytical features.

Secondly, if a company wants to make its operations stable and to be ready even when there is an unexpected increase of data, the best thing to do is to adopt self-regulating frameworks. These systems make use of machine learning to automatically increase the processing capacity whenever more data appears without a need for manual intervention (Floratou & Agrawal, 2017). This approach keeps performance stable and at the same time optimizes infrastructure costs.

Lastly, with regard to the problem of data heterogeneity, companies should incorporate a very powerful data integration and reasoning approach. High-performance frameworks such as Apache Spark or Flink should be adopted, as they are capable of ingesting and processing a variety of data formats efficiently (Hesse & Lorenz, 2015). In addition, integrating stream reasoning capabilities enables systems to perform real-time analytical reasoning, enabling them to derive insights from raw streaming data (Özbulak et al., 2025).

5.0 CONCLUSION

In conclusion, the present report points out that both batch and streaming data processing are vital and complementary in modern enterprise systems. The benefits of batch processing are that it is simple, reliable, and cost-effective, whereas streaming data processing allows instant analysis and makes quick decisions in a dynamic environment (Chen et al., 2014; Kleppmann, 2017). The analysis reveals that despite the great benefits of streaming architectures, the technology also presents several challenges, which include the complexity of the system, the heterogeneity of data, and system instability. Modern solutions (Kappa architecture and self-regulating systems) are the answer to these issues as they will make processing easier and scalable (Penka et al., 2021; Floratou and Agrawal, 2017).

Finally, organizations have to strike a middle ground by employing either of the paradigms depending on particular business requirements. Besides that, future research should be directed at further increasing automation, making the architecture less complex, and improving interoperability, which are key factors for achieving great scalability and intelligent enterprise data systems.

REFERENCES

- Benjelloun, S., Lakhrissi, Y., El Aissi, M. E. M., Elhaj Ben Ali, S., El Boushaki, A., Loukili, Y., & Chougrad, H. (2020). Big data processing: Batch-based processing and stream-based processing. *2020 IEEE Conference Proceedings*.
- Chen, M., Mao, S., & Liu, Y. (2014). Big data: A survey. *Mobile Networks and Applications*, 19(2), 171–209. <https://doi.org/10.1007/s11036-013-0489-0>
- Estuary. (2023). *Streaming data processing: Stream vs batch processing*[Image]. https://estuary.dev/static/95bfd47b6f3119035d2284a7c076fb5d/a4da0/02_Streaming_Data_Processing_Stream_Vs_Batch_Processing_7c1ac11dc8.png
- Gandomi, A., & Haider, M. (2015). Beyond the hype: Big data concepts, methods, and analytics. *International Journal of Information Management*, 35(2), 137–144. <https://doi.org/10.1016/j.ijinfomgt.2014.10.007>
- Golab, L., & Özsu, M. T. (2003). Issues in data stream management. *SIGMOD Record*, 32(2), 5-14.
- Hashem, I. A. T., Yaqoob, I., Anuar, N. B., Mokhtar, S., Gani, A., & Ullah Khan, S. (2015). The rise of “big data” on cloud computing: Review and open research issues. *Information Systems*, 47, 98–115. <https://doi.org/10.1016/j.is.2014.07.006>
- Hesse, G., & Lorenz, M. (2015). Conceptual survey on data stream processing systems. *2015 IEEE 21st International Conference on Parallel and Distributed Systems*.
- Floratou, A., & Agrawal, A. (2017). Self-regulating streaming systems: Challenges and opportunities. In *Proceedings of the International Workshop on Real-Time Business Intelligence and Analytics (BIRTE '17)* (pp. 1–5). <https://doi.org/10.1145/3129292.3129295>
- Kleppmann, M. (2017). *Designing data-intensive applications: The big ideas behind reliable, scalable, and maintainable systems*. O'Reilly Media.
- Koganti, V. K., Saini, R. K., Rajendra, G. Y., & Singh, P. N. (2025). High-performance streaming architectures for modern enterprise systems. *2025 International Conference on Sustainability, Innovation & Technology (ICSIT)*.

Özbulak, G. A. T., Manzo, G., Shrestha, Y. R., & Calbimonte, J. P. (2025). A comprehensive survey of stream reasoning and its integration with knowledge graphs. *Knowledge and Information Systems*, 67, 11225-11257.

Penka, J. B. N., Mahmoudi, S., & Debauche, O. (2021). A new Kappa architecture for IoT data management in smart farming. *Procedia Computer Science*, 191, 17-24.

Appendix / Logbook

Course: SECP3843 – Special Topic In Data Engineering

Semester: 2025/2026 Semester II

Topic: Streaming Data and Batch Processing

Group: 27

Members: Evelyn Goh Yuan Qi (A23CS0222), Joanne Ching Yin Xuan (A23CS0227)

1. Research and Writing Timeline

Date	Activity	Tools	Keyword	Details / Outcome
17 Mac 2026	Assignment briefing and topic selection	e-Learning, Google Sheets	“Enterprise Systems”, “batch processing vs streaming data,”	After the lecturer’s briefing, our group decided to choose “Streaming Data and Batch Processing.” Google Sheets used to record members and topics
18-21 Mac 2026	Research on data processing concepts	Google Scholar, IEEE Xplore, ChatGPT, e-Learning	“batch processing vs streaming data,” “advantages and challenges of streaming data,” “case studies”	Collected academic and industry resources related to streaming data and batch processing. Sources included journal articles, technical reports, and trusted online documentation.
22 Mac 2026	Research outcome discussion, task division, & Report planning	Spin the Wheel (online), Whatsapp, e-Learning	“technical report structure,” “task division,” “team collaboration”	Task distribution was completed on 20 Mac 2026, and the group agreed on deadlines for drafting, review, and final submission, and Spin the Wheel was used to assign tasks. Final task list: • Executive Summary – Evelyn • Introduction – Joanne • Literature Review – Evelyn • Analysis and Discussion –Joanne •Challenges /Future Trends – Evelyn • Recommendations –Evelyn • Conclusion – Joanne
23 Mac 2026	Introduction	Google Scholar, ChatGPT, Microsoft Word	“enterprise data processing,” “streaming data vs batch processing,” “report objective scope focus.”	Developed the introduction to provide background on streaming data and batch processing in an enterprise. Conducted basic research using Google Scholar. Used ChatGPT to help structure the flow of ideas. Prompt used: <i>Provide a research paper link related to streaming data and</i>

				<i>batch processing</i> . All content was verified and refined before finalizing in Microsoft Word.
25 Mac 2026	Background/ Literature review	Google Scholar, ChatGPT, Microsoft Word	"Batch processing", "Streaming data", "architecture", "Lambda", "Kappa"	Summarized the main findings on batch processing and streaming data from selected journal articles. Used ChatGPT with the prompt: <i>"Summarize important information about batch processing and streaming data from academic sources."</i> The AI output was paraphrased and reviewed against sources to ensure factual accuracy and citation consistency before inclusion in the report.
29 Mac 2026	Analysis and Discussion	Google Scholar, ChatGPT, Microsoft Word	"Streaming Data & Batch Processing work ", "case example", "advantages", "disadvantages", "challenges", "future trends"	A. Technology works or functions: Prompt used: "Give me a brief overview of how Streaming Data & Batch Processing work" B. Case examples or applications: Prompt used: "Provide real industry case examples of Streaming Data & Batch Processing work and provide me the resources."
30 Mac 2026	Analysis and Discussion	Google Scholar, ChatGPT, Microsoft Word	"advantages", "disadvantages", "challenges", "future trends"	C. Advantages and disadvantages: Prompt used: "Provide me with the resources that discuss the advantages and disadvantages of the system." D. Challenges and future trends: Prompt: <i>"Summarize the challenges and future trends of the transition from batch processing to stream processing."</i> All AI-generated summaries were cross-checked to ensure factual accuracy and consistency before inclusion in the final report.
1 Apr 2026	Recommendations	Microsoft Word, ChatGPT, Google Scholar	"batch processing", "stream processing", "recommendations"	Used ChatGPT with the prompt: <i>"Summarize key recommendations for successfully transitioning to stream processing."</i> Verified the output with research sources.

4 Apr 2026	Conclusion	Microsoft Word, ChatGPT	“batch processing”, “stream processing”, “enterprise systems”, “real-time analytics”, “future trends”	Summarized key findings on batch and streaming processing, their roles in enterprise systems, main challenges, and future improvements, such as automation and scalability. Prompt used: <i>“Summarize the key findings of batch and streaming data processing in enterprise systems and suggest future directions.”</i>
8 Apr 2026	AI verification & Create a detailed appendix/logbook section, including AI prompts and verification steps.	Grammarly, Turnitin, Microsoft Word	“AI verification”, “grammar check”, “logbook preparation”, “academic writing”, “report clarity.”	Used Grammarly to refine grammar and Turnitin to check originality and improve clarity. Used ChatGPT to structure and summarize the logbook format. All sections were manually reviewed before inclusion in the final report.
9 Apr 2026	Editing and final review, Presentation slide	Microsoft Word, Grammarly, Turnitin, Power Point	“APA citation”, “report formatting”, “final editing”, “proofreading”, “submission.”	Check all sections, formatted headings and tables, reviewed APA citations, and performed final grammar and originality checks. Exported and finalized the report in PDF format for submission. A PowerPoint slide is prepared for presentation.

2. Verification Steps

Each section of the report was thoroughly verified to ensure accuracy and originality. All AI-generated summaries, tables, or paraphrased content were checked against the original journal sources before inclusion in the final document. No AI-generated text was used without manual editing and review to maintain academic integrity. Every reference was cross-checked through DOI links and formatted according to the APA 7th edition guidelines. Finally, Grammarly was used for grammar and clarity improvement, while Turnitin was employed to verify originality and ensure that the final report met academic standards.